Employee & Department SQL Project

1. Top 5 Highest Salary Employees

SQL Query:

```
SELECT *
FROM Employee
ORDER BY salary DESC
LIMIT 5;
```

Expected Output:

Output:
Neha 95000
Sneha 85000
Karan 80000
Priya 75000
Rohit 70000

2. Department Wise Employee Count

SQL Query:

```
SELECT d.dept_name, COUNT(e.emp_id) AS employee_count
FROM Department d
JOIN Employee e ON d.dept_id=e.dept_id
GROUP BY d.dept_name;
```

Expected Output:

Output:
HR = 3
IT = 3
Finance = 2
Marketing = 2

3. Second Highest Salary

SQL Query:

```
SELECT MAX(salary)
FROM Employee
WHERE salary < (SELECT MAX(salary) FROM Employee);
```

Expected Output:

Output: 85000

4. Salary Greater Than Department Average

SQL Query:

```
SELECT e.*
FROM Employee e
JOIN (
 SELECT dept_id, AVG(salary) avg_salary
```

```
 FROM Employee
 GROUP BY dept_id
) a ON e.dept_id=a.dept_id
WHERE e.salary > a.avg_salary;
```

Expected Output:

Output: Priya, Karan, Sneha, Neha, Rohit

5. Inner Join

SQL Query:

```
SELECT e.emp_id,e.emp_name,d.dept_name
FROM Employee e
INNER JOIN Department d
ON e.dept_id=d.dept_id;
```

Expected Output:

Output: Employee details with matching department names

6. Left Join

SQL Query:

```
SELECT e.emp_id,e.emp_name,d.dept_name
FROM Employee e
LEFT JOIN Department d
ON e.dept_id=d.dept_id;
```

Expected Output:

Output: All employees with department information

7. Group By Having

SQL Query:

```
SELECT dept_id,COUNT(*)
FROM Employee
GROUP BY dept_id
HAVING COUNT(*) > 2;
```

Expected Output:

Output: HR, IT

8. Employees Hired In Last 6 Months

SQL Query:

```
SELECT *
FROM Employee
WHERE hire_date >= DATE_SUB(CURDATE(), INTERVAL 6 MONTH);
```

Expected Output:

Output: Amit, Rahul, Vikas, Rohit, Pooja, Anjali

9. Find Duplicate Records

SQL Query:

```
SELECT emp_name,salary,dept_id,COUNT(*)
FROM Employee
GROUP BY emp_name,salary,dept_id
HAVING COUNT(*) > 1;
```

Expected Output:

Output: Amit, Priya

10. Remove Duplicate Records

SQL Query:

```
WITH DuplicateRecords AS (
 SELECT *,
 ROW_NUMBER() OVER(
 PARTITION BY emp_name,salary,dept_id
 ORDER BY emp_id) rn
 FROM Employee
)
DELETE FROM DuplicateRecords
WHERE rn > 1;
```

Expected Output:

Output: Duplicate rows removed